

AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the LEOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

- I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.
- II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.
- III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

On verification. The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganizes the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 128	Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater–Nichols.	
	Case 137	U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.	
124	USS Fitzgerald & USS John S. McCain	2017	T-K-N
126	F-35 Sustainment & the Maintainer Capability Gap	ongoing	T-K-D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T-H-K
128	Operation Eagle Claw	1980	T-K
129	Patriot Missile / Dhahran	1991	D-H-K
130	USS Vincennes & Iran Air Flight 655	1988	H-T
131	Stanislav Petrov / 1983 False Alert	1983	H-T
133	Mark 14 Torpedo Failures	1941 – 1943	T-K-G
134	V-22 Osprey	1991 – present	T-H-N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G-K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T-K-N
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G-K

139	GIFT and the Adoption Gap	2012 – present	K·G·N
140	Navy Surface Warfare Readiness Reform	2018 – present	T·K·N
141	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D

AVIATION

STANDOUT FAILURE

Case 97

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS

Case 117

Crew Resource Management & CAST (1981–). Authority gradients named and engineered into the cockpit, then a closed-loop industry data architecture sixteen years later. The 83% fatality-risk reduction CAST reports is a portfolio result; CRM's separate contribution is not isolated in any published estimate.

96	AeroPerú Flight 603	1996	H·T
97	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
102	Air France Flight 447	2009	T·H
103	Kegworth / British Midland 92	1989	T·H·K
104	Asiana Airlines Flight 214	2013	T·H
105	Helios Airways Flight 522	2005	T·H
106	TransAsia Airways Flight 235	2015	T·H
107	Eastern Air Lines Flight 401	1972	H·T
109	Colgan Air Flight 3407	2009	T
110	Atlas Air Flight 3591	2019	T·K
112	NASA Saturn V Documentation	1972 – present	K
117	Crew Resource Management & CAST	1981 – present	T·H·N
118	Korean Air Safety Transformation	2000 – present	T·N
119	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
120	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
121	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson Version 7.1)	1981 – 2008 (TCAS II)	H·K·G
122	Singapore Airlines Safety Transformation	1980s – present	T·N

132 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap 2008 – 2012 **H-K-N**

185 Air Canada Chatbot Liability — Delegation Without Revocation 2022 – 2024 **D-K-N**

HEALTHCARE

STANDOUT FAILURE

Case 2

EHR / CPOE Implementation (2005–). Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS

Case 23

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

1 Therac-25 1985 – 1987 **H-D-G**

2 EHR / CPOE Implementation 2005 – present **H-D-G**

4 Mid Staffordshire NHS Foundation Trust 2005 – 2009 **G-N-K**

5 Epic Sepsis Model 2017 – 2021 **D-G-N**

8 Medical Errors as Systemic Failure 1999 – present **T-H-N-K-G**

9 Vioxx Withdrawal 1999 – 2004 **G-D**

11 California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010 **G-K**

12 ACGME 80-Hour Resident Duty-Hour Reform 2003–2017 **T-K-N**

13 Implementation Science in Healthcare — The 17-Year Gap ongoing **K-G-N**

19 Keystone ICU / Pronovost Checklist 2004 – present **T-N**

20 TREWS / Sepsis Watch 2018 – 2022 **H-K-G**

21 Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff 2024 – 2025 **H-K-N**

22 ChatGPT in Healthcare — Hallucination Cases 2023 – present **H-D**

23 WHO Surgical Safety Checklist 2008 – present **T-N**

24 Bristol Heart Babies Reform 1984 – present **G-N**

32 Annual-Screening UI Redesign at University of Missouri Health Care 2020 **H-N**

33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal 2019 – 2024 **T-D-N**

34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case 2022 – 2024 **T-K-D**

35 Radiology AI Miscalibration 2018 – present **H-K-D**

36	AlphaFold — Protein Structure Prediction	2020 – present	K·G
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback	2020	K·G
39	TeamSTEPPS	2006 – present	T·N
176	Tylenol Recall	1982	G·N

EDUCATION AT SCALE

STANDOUT FAILURE

Case 54

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS

Case 80

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45	Tennessee Voluntary Pre-K Study	2009–2018	G·D
46	Algorithmic Bias in Educational Predictive Analytics	ongoing	G·H·D
51	Atlanta Public Schools Cheating Scandal	2009 – 2015	G·N
53	inBloom	2014	G
54	Summit Learning / Personalized Learning Rollout	2014–2019	G·T·K
67	Cognitive Tutor / Carnegie Learning	1990s – present	T
80	Georgia State University Predictive Analytics	2012 – present	T·K
142	xAPI / Total Learning Architecture — Interoperability Gap	ongoing	K·G
205	The Discipline We Build Next	ongoing	T·K·N·G·H·D

ENERGY

STANDOUT FAILURE

Case 166

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT SUCCESS

Case 175

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES Level 4-or-above accident at a U.S. commercial reactor since.

146	Northeast Blackout	2003	H·K
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160	Davis-Besse Reactor Head Corrosion	2002	N-K-G
161	Texas City BP Refinery Explosion	2005	N-T-K-G
162	Upper Big Branch Mine Explosion	2010	N-G-K
163	Deepwater Horizon	2010	T-N-K
165	Camp Fire / PG&E	2018	G-N-K
166	Three Mile Island	1979	T-H
169	Fukushima Daiichi	2011	N-G-K
175	INPO and the Nuclear Academy	1979 – present	T-K-G

GOVERNMENT

STANDOUT FAILURE

Case 191

Australia Robodebt (2016–2023). An income-averaging algorithm that reversed the burden of proof onto welfare recipients, raising some 470,000 wrongful debts. The Royal Commission found the scheme unlawful and attributed deaths to its operation — circumstantially, not as a finding of individual causation.

NOTE

Case 203

The reforms that follow these failures — Royal Commissions, statutory restructures — are documented inside the narratives rather than as separate cases. The government dataset's paired-intervention success is NYC Local Law 144 (Case 203), filed under its own primary domain.

7	VA Wait-Time Scandal	2014	G-K-N
180	Healthcare.gov Launch	2013	K-T-G
193	Bernard Madoff / SEC Failures	1992 – 2008	G-K-N
197	Predictive Policing — PredPol	2011 – present	G-H-D

INDUSTRIAL

STANDOUT FAILURE

Case 159

Bhopal (1984). Training, knowledge, normalization of deviance, and governance all collapsed in the same plant. Thousands died within hours; total-death estimates run to 15,000–20,000, with roughly half a million exposed. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS

Case 155

Toyota Production System / Andon Cord (1950s–). Any operator can signal a defect at the source. Under the fixed-position stop system most pulls never halt the line — the smallest authority-architecture intervention in the dataset, and the most durably copied.

144	Kodak Digital Camera Stagnation	1975–2012	D-K-N
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147	Takata Airbag Inflators	2008 – 2026	D-G
149	Volkswagen Dieselgate	2015	D-G
151	GM Ignition Switch	2002 – 2014	D-G
155	Toyota Production System / Andon Cord	1950s – present	N-G
159	Bhopal	1984	T-K-N-G
164	Grenfell Tower	2017	G-T-K-N

TECHNOLOGY

STANDOUT FAILURE

Case 184

UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. More than 900 wrongful convictions, 236 imprisoned, and 13 suicides the inquiry could not rule out as Horizon-caused — the longest-running operator-feedback failure in the dataset.

NOTE

Case 36

The technology dataset's success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

143	Knight Capital Trading Loss	2012	D-K
148	Wells Fargo Fake Accounts	2011 – 2016	G-N
152	TSB Bank IT Migration	2018	H-G
153	Equifax Data Breach	2017	G-K
157	AI-Augmented Coding Tools	2021 – present	T-H
172	CrowdStrike Falcon Outage	2024	D-K-G
184	UK Post Office Horizon Scandal	1999 – 2015	G-H-K

AI IN EDUCATION

88	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T-K-N
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K-12 EDUCATION

48	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G-K-N
60	Houston EVAAS — Value-Added Teacher Evaluation on Trial	2011–2017	G
61	Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap	1997–2023	K-T
62	Gates Intensive Partnerships — Measurement Without a Coupled Lever	2009–2018	G-K

63	LAUSD iPad Program — The Common Core Technology Project	2013–2015	D·H·G
66	Growth–Mindset National Experiment — Heterogeneous Effects	2019	D·N·K
75	Chen et al. — Rural China AI Devices and the Equity–Direction Finding	2025	T·K·D
95	PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop	1997–2024	T·G

K–12 MATHEMATICS

72	ASSISTments — National Replication and Long–Term Follow–Through	2014 – present	T·K·D
84	Cognitive Tutor Algebra I at Scale — Year–One Null, Year–Two Positive	2007 – 2014	T·K·D

K–12 SCIENCE

74	Zhang/Scardamalia — Knowledge Building Across Three Cohorts	2009	T·K·D
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AIR TRAFFIC MANAGEMENT

116	Eurocat ATM Pilot Modernization — Small–Tier Verification as the Gateway to Big–Tier Change	2005	G·K·H
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ANESTHESIOLOGY

27	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
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AUTONOMOUS SYSTEMS

STANDOUT FAILURE

Case 183

Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE

Case 205

This key’s cases are all failures and frontiers; the paired-intervention successes — Waymo’s safety case (199) and the CPUC permit regime (200) — are filed under autonomous vehicles. See Case 205 (**The Discipline We Build Next**) for the open question this volume closes on.

183	Uber ATG / Tempe Fatality	2018	T·N·G·H
195	Tesla Autopilot — Recurring Fatalities	2016 – present	T·N·G·H

AUTONOMOUS VEHICLES

190	Cruise’s Partial Disclosure — How Disclosure Posture Decides Deployment	2023 – 2024	G·K·N
199	Waymo’s Safety Case Framework — Governance Objection Dissolved by Designed Artifact	2023 – 2026	G·K·N

200	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver 2020 – 2026	G-K-D
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AVIATION INFRASTRUCTURE

115	FAA NextGen and the ADS-B Out Transition 2003 – 2020	G-D-K
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AVIATION SAFETY

114	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s	G-D-K
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123	FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike 1992 – 2000s	G-K-N
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CLINICAL CARE

15	I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer 2014	H-K-N
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29	Bar-Code Medication Administration — Cue at the Point of Care 2010	H-K-D
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CLINICAL MEDICINE

6	Racial Bias in Pain Assessment — The False-Belief Mechanism 2016	T-K-N
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25	Removing the Race Coefficient from eGFR 2021	D-G-N
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CLINICAL/TRANSLATIONAL RESEARCH

40	Team Science Training for Clinical and Translational Scientists 2020 – 2022	K-N
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CORPORATE L&D

65	Training Transfer — The Gap Between Learning and Doing 2010	K-N
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70	High-Impact Learning System — Engineering the Environment, Not Just the Event 2001 – present	K-N
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79	The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present	K-N
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83	Brinkerhoff Success Case Method — Tails as the Evaluation Instrument 2003 – present	K-N
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CRIMINAL JUSTICE

187	COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate 2014 – 2018	D-K-N
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DIGITAL IDENTITY

201	Aadhaar Exclusion — Biometric Welfare Delegation and Its Judicial Reckoning in India 2017 – 2025	G-N-H
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E-GOVERNMENT

194	Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) 2001 – present	G-K-N
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ED-TECH

47 Algorithmic Bias in Automated Exam Proctoring 2022 **D·N·K**

EDUCATION (NORWAY)

92 Norway's National Expert Commission on Learning Analytics 2021 – 2023 **G·K·N**

EDUCATION AT SCALE

50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2023 **D·K·N**

69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T·D**

EMERGENCY MANAGEMENT

167 Hurricane Katrina — The FEMA/DHS Response Failure 2005 **G·K**

171 Hurricane Maria — Puerto Rico Response and Logistics Failure 2017 – 2018 **G·N**

FINANCE

186 Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity 2009 – 2022 **D·G·N**

196 Fintech Lending Fairness Audit — When Including Race Reduces Disparity 2025 **D·G·N**

FINANCIAL SERVICES

192 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021 **D·K·N**

GENERAL AVIATION

100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010 **H·N·K**

GLOBAL HEALTH

18 PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023 **K·N·D**

43 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018 **H·N**

170 West Africa Ebola — The Delayed International Response 2014 – 2016 **G·N**

GOVERNMENT

49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020 **D·K·N**

191 Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict 2016 – 2023 **D·G·N**

203 NYC Local Law 144 — Bias Audits as Governance Artifact 2023 – present **D·G·K**

GOVERNMENT/WELFARE (NETHERLANDS)

189 Dutch SyRI — Welfare–Fraud Risk Scoring Halted on Rights Grounds 2014 – 2020 **D·G·N**

HEALTH PROFESSIONS EDUCATION

28 Interprofessional Education and the Evidence Gap 2013 – 2015 **K·N**

HEALTHCARE/ONCOLOGY

3 Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018 **D·K·N**

HIGHER EDUCATION

55 Enrollment–Algorithm Yield Optimization Across U.S. Higher Education 2010s – present **T·K·N**
(Brookings synthesis 2021)

57 GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022 **D·K·N**

58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024 **D·K·N**

85 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023 **T·K·D**

86 Gándara — Detecting and Mitigating Algorithmic Bias in College Student–Success Prediction 2024 **D·K·N**

90 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025 **T·K·N**

HIGHER EDUCATION (AFRICA)

89 Data Privacy and Learning Analytics on the African Continent 2022 **K·G·N**

HIGHER EDUCATION (BRAZIL)

82 MMALA — A Maturity Model for Responsible Learning–Analytics Adoption (Brazil) 2024 **K·N**

HIGHER EDUCATION (LATIN AMERICA)

91 LALA — Building Learning–Analytics Governance Capacity Across Latin America 2017 – 2020 **K·N**

HIGHER ED (UK)

81 Open University 'Ethical Use of Student Data' and OU Analyse 2014 – 2025 **G·K·N**

HIGHER-ED ANALYTICS

52 Purdue Course Signals — The Reverse–Causality Retention Claim 2012 – 2013 **D·K·N**

HOSPITAL PHARMACY

42 Australian Hospital–Pharmacy Technician Role Redesign 2016 **D·N·H**

HUMANITARIAN RELIEF

-
- | | | |
|------------|--|----------|
| 178 | The UN Cluster Approach — Engineering Humanitarian Coordination After the Tsunami
2005 – 2020 | G |
|------------|--|----------|
-

IMPLEMENTATION SCIENCE

-
- | | | |
|-----------|---|------------|
| 41 | Translational-Workforce Training — Stated Goals Outrunning Named Practice 2020s | K·N |
|-----------|---|------------|
-

LEARNING ANALYTICS

-
- | | | |
|-----------|--|--------------|
| 73 | The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension
2016 – 2025 | T·K·D |
|-----------|--|--------------|
-
- | | | |
|-----------|---|--------------|
| 87 | Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models 2021 – 2022 | D·K·N |
|-----------|---|--------------|
-
- | | | |
|-----------|--|------------|
| 94 | Learning Analytics on the African Continent — A Scoping Review as the Present-State Map 2022 | K·N |
|-----------|--|------------|
-

MEDICAL DEVICES

-
- | | | |
|-----------|---|--------------|
| 26 | Pulse Oximetry Across Skin Tones 1990 – 2020 – 2025 | D·G·N |
|-----------|---|--------------|
-

MEDICAL EDUCATION

-
- | | | |
|-----------|--|------------|
| 14 | Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA 2009 – 2020s | T·K |
|-----------|--|------------|
-
- | | | |
|-----------|---|--------------|
| 17 | Spaced Education RCTs in Medical Training 2007 – 2009 | H·K·D |
|-----------|---|--------------|
-

MEDICAL-DEVICE REGULATION

-
- | | | |
|-----------|---|------------|
| 44 | Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD 2014 – present | G·N |
|-----------|---|------------|
-

MULTIMODAL LEARNING ANALYTICS

-
- | | | |
|-----------|---|--------------|
| 76 | Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account 2023 | T·K·N |
|-----------|---|--------------|
-

NAVAL ENGINEERING

-
- | | | |
|------------|--|--------------|
| 173 | Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable 1963 – present | G·K·H |
|------------|--|--------------|
-

NEUROSCIENCE

-
- | | | |
|------------|---|------------|
| 181 | EU Human Brain Project — Top-Down Vision That Unraveled 2013 – 2023 | G·N |
|------------|---|------------|
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- | | | |
|------------|--|--------------|
| 198 | Launching the BRAIN Initiative — Governance of a Grand Challenge 2011 – 2015 – present | G·K·N |
|------------|--|--------------|
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NEUROSCIENCE/CONNECTOMICS

-
- | | | |
|-----------|---|--------------|
| 78 | CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale 2017 – present | H·K·N |
|-----------|---|--------------|
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NUCLEAR ENGINEERING

156	INL / LWRs Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present	D·H·K
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NUCLEAR POWER

154	INL Turbine-Control Upgrade — Verification and Validation as the Cutover Deliverable 2014	G·K·H
-----	--	-------

NURSING EDUCATION

16	NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014	G·K·D
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PHARMA SAFETY

38	iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms — 2011	G·D·N
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PUBLIC HEALTH

179	The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed 2020 – 2023	G·K
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RAIL TRANSPORT

177	CIRAS — Confidential Incident Reporting for UK Rail 1996 – present	G·K·N
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SOFTWARE ENGINEERING EDUCATION

71	Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development 2025	K·N
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SOFTWARE SUSTAINMENT

174	Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed 1996 – 2000	G·D·K
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SPACE / AEROSPACE

STANDOUT FAILURE	Case 111	Challenger & Columbia (1986 / 2003). Normalisation of deviance across seventeen years and two crews — the sociologist Diane Vaughan’s term for a culture that accepted flying with flaws once problems were defined as normal and routine, and the culture the CAIB found still intact in 2003.
NOTE	Case 137	No paired-intervention success in the space dataset — Part III’s aviation half carries seven. See Defense (Rickover, Case 137) for the safety-engineering counterpart human spaceflight did not build.

98	Mars Climate Orbiter — Unit Mismatch 1999	D·K
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101	Ariane 5 Flight 501 1996	D·K·H
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108	NASA EVA-23 — Water Intrusion Inside a Spacesuit	2013	H·N·D
111	Challenger & Columbia	1986 / 2003	N·K·G
113	Boeing Starliner	2019 – 2025	K·D

SURGERY

30	Surgical Skill Video Peer-Rating Predicts Complications	2013	H·D·K
31	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units	2006 – 2016	T·K·H

TECHNOLOGY

182	Amazon Hiring AI — Trained Bias, Deprecated 2017	2014 – 2018	D·K·N
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TUTORING

77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation	2024	T·K·D
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WORKFORCE DEV

68	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model	2017 – 2023 (six cycles)	T·K
93	Singapore SkillsFuture — National Workforce Capability at Scale	2015 – present	G·K·D

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the concentration, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

36 cases

6	Racial Bias in Pain Assessment — The False-Belief Mechanism
8	Medical Errors as Systemic Failure
13	Implementation Science in Healthcare — The 17-Year Gap
36	AlphaFold — Protein Structure Prediction
39	TeamSTEPPS
53	inBloom
63	LAUSD iPad Program — The Common Core Technology Project
67	Cognitive Tutor / Carnegie Learning
68	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model
80	Georgia State University Predictive Analytics
97	Boeing 737 MAX / MCAS
102	Air France Flight 447
111	Challenger & Columbia
112	NASA Saturn V Documentation
114	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule
115	FAA NextGen and the ADS-B Out Transition
116	Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change
117	Crew Resource Management & CAST
124	USS Fitzgerald & USS John S. McCain
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

- 139 GIFT and the Adoption Gap
- 141 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks
- 154 INL Turbine–Control Upgrade — Verification and Validation as the Cutover Deliverable
- 156 INL / LWRS Control–Room Modernization — Sustainment Research for an Aging Fleet
- 166 Three Mile Island
- 167 Hurricane Katrina — The FEMA/DHS Response Failure
- 171 Hurricane Maria — Puerto Rico Response and Logistics Failure
- 173 Navy SUBSAFE — Requirements as a Non–Negotiable Sustainment Deliverable
- 174 Y2K Remediation — The Aging–System Transition That Worked Because It Was Believed
- 175 INPO and the Nuclear Academy
- 178 The UN Cluster Approach — Engineering Humanitarian Coordination After the Tsunami
- 180 Healthcare.gov Launch
- 181 EU Human Brain Project — Top–Down Vision That Unraveled
- 194 Estonia X–Road — Continuous Migration as a Governance Pattern (and the No–Legacy Paradox)
- 198 Launching the BRAIN Initiative — Governance of a Grand Challenge
- 205 The Discipline We Build Next

LEN 2

53 cases

- 1 Therac–25
- 2 EHR / CPOE Implementation
- 5 Epic Sepsis Model
- 14 Barsuk SBML — Simulation–Based Mastery Learning Dissemination from Northwestern to the VA
- 15 I–PASS Handoff Bundle — Structuring the Human–to–Human Transfer
- 16 NCSBN National Simulation Study — Licensing the 50% Substitution Rule
- 17 Spaced Education RCTs in Medical Training
- 18 PEPFAR HIV Training Across 16 Sub–Saharan African Countries — Modality Comparison Under Disruption
- 20 TREWS / Sepsis Watch
- 22 ChatGPT in Healthcare — Hallucination Cases
- 27 Anesthesia Monitoring Standards and the APSF — The Mortality Decline
- 29 Bar–Code Medication Administration — Cue at the Point of Care
- 30 Surgical Skill Video Peer–Rating Predicts Complications

- 131 Stanislav Petrov / 1983 False Alert
- 141 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks
- 144 Kodak Digital Camera Stagnation
- 146 Northeast Blackout
- 155 Toyota Production System / Andon Cord
- 157 AI-Augmented Coding Tools
- 172 CrowdStrike Falcon Outage
- 183 Uber ATG / Tempe Fatality
- 184 UK Post Office Horizon Scandal
- 195 Tesla Autopilot — Recurring Fatalities

LEN 3

47 cases

- 4 Mid Staffordshire NHS Foundation Trust
- 17 Spaced Education RCTs in Medical Training
- 24 Bristol Heart Babies Reform
- 32 Annual-Screening UI Redesign at University of Missouri Health Care
- 33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal
- 34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case
- 37 DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback
- 39 TeamSTEPPS
- 49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days
- 50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction
- 55 Enrollment-Algorithm Yield Optimization Across U.S. Higher Education
- 57 GAO Online Program Manager Oversight Gap (GAO-22-104463)
- 58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)
- 72 ASSISTments — National Replication and Long-Term Follow-Through
- 73 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension
- 74 Zhang/Scardamalia — Knowledge Building Across Three Cohorts
- 75 Chen et al. — Rural China AI Devices and the Equity-Direction Finding
- 84 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive
- 85 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

86	Gándara — Detecting and Mitigating Algorithmic Bias in College Student–Success Prediction
87	Yu / Lee / Kizilcec — Protected Attributes in Learning–Analytics Models
88	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models
90	Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions
108	NASA EVA–23 — Water Intrusion Inside a Spacesuit
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124	USS Fitzgerald & USS John S. McCain
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132	F–22 OBOGS Hypoxia — The Sustainment–Era Instrumentation Gap
134	V–22 Osprey
135	9/11 Intelligence Sharing Failures
137	U.S. Nuclear Navy / Rickover Training Model
144	Kodak Digital Camera Stagnation
154	INL Turbine–Control Upgrade — Verification and Validation as the Cutover Deliverable
159	Bhopal
161	Texas City BP Refinery Explosion
163	Deepwater Horizon
164	Grenfell Tower
169	Fukushima Daiichi
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182	Amazon Hiring AI — Trained Bias, Deprecated 2017
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-
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 - 70 High-Impact Learning System — Engineering the Environment, Not Just the Event
 - 79 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops
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 - 81 Open University 'Ethical Use of Student Data' and OU Analyse
 - 82 MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)
 - 83 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument
 - 89 Data Privacy and Learning Analytics on the African Continent
 - 91 LALA — Building Learning-Analytics Governance Capacity Across Latin America
 - 92 Norway's National Expert Commission on Learning Analytics
 - 93 Singapore SkillsFuture — National Workforce Capability at Scale
 - 94 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map
 - 95 PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop
 - 109 Colgan Air Flight 3407
 - 110 Atlas Air Flight 3591
 - 117 Crew Resource Management & CAST
 - 119 Aviation Safety Reporting System (ASRS)
 - 123 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike
 - 140 Navy Surface Warfare Readiness Reform
 - 142 xAPI / Total Learning Architecture — Interoperability Gap
 - 147 Takata Airbag Inflators
 - 148 Wells Fargo Fake Accounts
 - 149 Volkswagen Dieselgate
 - 151 GM Ignition Switch
 - 161 Texas City BP Refinery Explosion
 - 162 Upper Big Branch Mine Explosion
 - 177 CIRAS — Confidential Incident Reporting for UK Rail
 - 178 The UN Cluster Approach — Engineering Humanitarian Coordination After the Tsunami
 - 179 The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed
-

186	Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity
189	Dutch SyRI — Welfare–Fraud Risk Scoring Halted on Rights Grounds
190	Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment
193	Bernard Madoff / SEC Failures
196	Fintech Lending Fairness Audit — When Including Race Reduces Disparity
199	Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact
200	CPUC AV Passenger–Service Permits — Conditions as a Designed Objection–Dissolver

LEN 5

84 cases

1	Therac-25
3	Watson for Oncology — Delegation Marketed Ahead of Capability
11	California Nurse Staffing Ratios — Legislating a Capability Requirement
12	ACGME 80–Hour Resident Duty–Hour Reform
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 - 62 Gates Intensive Partnerships — Measurement Without a Coupled Lever

 - 63 LAUSD iPad Program — The Common Core Technology Project

 - 66 Growth-Mindset National Experiment — Heterogeneous Effects

 - 68 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

 - 69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

 - 76 Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account

 - 77 Hybrid Human-AI Tutoring — Augmentation, Not Delegation

 - 78 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

 - 88 LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models

 - 90 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions

 - 95 PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop

 - 96 AeroPerú Flight 603

 - 97 Boeing 737 MAX / MCAS

 - 98 Mars Climate Orbiter — Unit Mismatch

 - 100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training

 - 101 Ariane 5 Flight 501

 - 102 Air France Flight 447

 - 103 Kegworth / British Midland 92

 - 104 Asiana Airlines Flight 214

 - 105 Helios Airways Flight 522

 - 106 TransAsia Airways Flight 235

 - 107 Eastern Air Lines Flight 401

 - 108 NASA EVA-23 — Water Intrusion Inside a Spacesuit

 - 109 Colgan Air Flight 3407

 - 113 Boeing Starliner

 - 120 EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation

 - 121 TCAS — Coordinated Collision Avoidance and the Überlingen Lesson

-
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-
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-
- 127 Military Fratricide — Desert Storm to Afghanistan
-
- 128 Operation Eagle Claw
-
- 129 Patriot Missile / Dhahran
-
- 130 USS Vincennes & Iran Air Flight 655
-
- 132 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap
-
- 134 V-22 Osprey
-
- 137 U.S. Nuclear Navy / Rickover Training Model
-
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-
- 140 Navy Surface Warfare Readiness Reform
-
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-
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-
- 153 Equifax Data Breach
-
- 159 Bhopal
-
- 163 Deepwater Horizon
-
- 166 Three Mile Island
-
- 167 Hurricane Katrina — The FEMA/DHS Response Failure
-
- 170 West Africa Ebola — The Delayed International Response
-
- 171 Hurricane Maria — Puerto Rico Response and Logistics Failure
-
- 172 CrowdStrike Falcon Outage
-
- 173 Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable
-
- 178 The UN Cluster Approach — Engineering Humanitarian Coordination After the Tsunami
-
- 180 Healthcare.gov Launch
-
- 182 Amazon Hiring AI — Trained Bias, Deprecated 2017
-
- 185 Air Canada Chatbot Liability — Delegation Without Revocation
-
- 187 COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate
-
- 191 Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict
-
- 192 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model
-
- 194 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)
-

201 Aadhaar Exclusion — Biometric Welfare Delegation and Its Judicial Reckoning in India

203 NYC Local Law 144 — Bias Audits as Governance Artifact

LEN 6

18 cases

8 Medical Errors as Systemic Failure

13 Implementation Science in Healthcare — The 17-Year Gap

53 inBloom

61 Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap

86 Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

87 Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

113 Boeing Starliner

116 Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

129 Patriot Missile / Dhahran

139 GIFT and the Adoption Gap

142 xAPI / Total Learning Architecture — Interoperability Gap

151 GM Ignition Switch

154 INL Turbine-Control Upgrade — Verification and Validation as the Cutover Deliverable

176 Tylenol Recall

180 Healthcare.gov Launch

193 Bernard Madoff / SEC Failures

194 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

195 Tesla Autopilot — Recurring Fatalities

LEN 7

102 cases

1 Therac-25

2 EHR / CPOE Implementation

3 Watson for Oncology — Delegation Marketed Ahead of Capability

4 Mid Staffordshire NHS Foundation Trust

5 Epic Sepsis Model

6 Racial Bias in Pain Assessment — The False-Belief Mechanism

7 VA Wait-Time Scandal

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|----|--|
| 11 | California Nurse Staffing Ratios — Legislating a Capability Requirement |
| 14 | Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA |
| 15 | I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer |
| 16 | NCSBN National Simulation Study — Licensing the 50% Substitution Rule |
| 18 | PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption |
| 20 | TREWS / Sepsis Watch |
| 21 | Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff |
| 22 | ChatGPT in Healthcare — Hallucination Cases |
| 24 | Bristol Heart Babies Reform |
| 25 | Removing the Race Coefficient from eGFR |
| 26 | Pulse Oximetry Across Skin Tones |
| 27 | Anesthesia Monitoring Standards and the APSF — The Mortality Decline |
| 28 | Interprofessional Education and the Evidence Gap |
| 29 | Bar-Code Medication Administration — Cue at the Point of Care |
| 30 | Surgical Skill Video Peer-Rating Predicts Complications |
| 31 | Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units |
| 35 | Radiology AI Miscalibration |
| 36 | AlphaFold — Protein Structure Prediction |
| 38 | iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms |
| 40 | Team Science Training for Clinical and Translational Scientists |
| 41 | Translational-Workforce Training — Stated Goals Outrunning Named Practice |
| 43 | Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface |
| 44 | Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD |
| 45 | Tennessee Voluntary Pre-K Study |
| 46 | Algorithmic Bias in Educational Predictive Analytics |
| 48 | School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism |
| 51 | Atlanta Public Schools Cheating Scandal |
| 53 | inBloom |
| 54 | Summit Learning / Personalized Learning Rollout |
| 60 | Houston EVAAS — Value-Added Teacher Evaluation on Trial |

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|-----|--|
| 65 | Training Transfer — The Gap Between Learning and Doing |
| 70 | High-Impact Learning System — Engineering the Environment, Not Just the Event |
| 71 | Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development |
| 72 | ASSISTments — National Replication and Long-Term Follow-Through |
| 73 | The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension |
| 74 | Zhang/Scardamalia — Knowledge Building Across Three Cohorts |
| 75 | Chen et al. — Rural China AI Devices and the Equity-Direction Finding |
| 76 | Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account |
| 77 | Hybrid Human-AI Tutoring — Augmentation, Not Delegation |
| 78 | CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale |
| 79 | The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops |
| 80 | Georgia State University Predictive Analytics |
| 81 | Open University 'Ethical Use of Student Data' and OU Analyse |
| 82 | MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil) |
| 83 | Brinkerhoff Success Case Method — Tails as the Evaluation Instrument |
| 84 | Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive |
| 85 | OU Analyse — Predictive Learning Analytics and Teacher Use at Scale |
| 88 | LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models |
| 89 | Data Privacy and Learning Analytics on the African Continent |
| 91 | LALA — Building Learning-Analytics Governance Capacity Across Latin America |
| 92 | Norway's National Expert Commission on Learning Analytics |
| 94 | Learning Analytics on the African Continent — A Scoping Review as the Present-State Map |
| 100 | Glass-Cockpit Transition in Light Aircraft — Technology Outran Training |
| 108 | NASA EVA-23 — Water Intrusion Inside a Spacesuit |
| 111 | Challenger & Columbia |
| 114 | FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule |
| 115 | FAA NextGen and the ADS-B Out Transition |
| 120 | EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation |
| 121 | TCAS — Coordinated Collision Avoidance and the Überlingen Lesson |
| 123 | FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike |

-
- 132 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap
-
- 133 Mark 14 Torpedo Failures
-
- 135 9/11 Intelligence Sharing Failures
-
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-
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-
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-
- 149 Volkswagen Dieselgate
-
- 151 GM Ignition Switch
-
- 152 TSB Bank IT Migration
-
- 153 Equifax Data Breach
-
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-
- 159 Bhopal
-
- 160 Davis-Besse Reactor Head Corrosion
-
- 161 Texas City BP Refinery Explosion
-
- 162 Upper Big Branch Mine Explosion
-
- 164 Grenfell Tower
-
- 165 Camp Fire / PG&E
-
- 167 Hurricane Katrina — The FEMA/DHS Response Failure
-
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-
- 171 Hurricane Maria — Puerto Rico Response and Logistics Failure
-
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-
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-
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-
- 180 Healthcare.gov Launch
-
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-
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-
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-
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-
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-

195 Tesla Autopilot — Recurring Fatalities

196 Fintech Lending Fairness Audit — When Including Race Reduces Disparity

197 Predictive Policing — PredPol

198 Launching the BRAIN Initiative — Governance of a Grand Challenge

LEN 8

90 cases

7 VA Wait-Time Scandal

11 California Nurse Staffing Ratios — Legislating a Capability Requirement

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-
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 - 98 Mars Climate Orbiter — Unit Mismatch

 - 101 Ariane 5 Flight 501

 - 103 Kegworth / British Midland 92

 - 109 Colgan Air Flight 3407

 - 110 Atlas Air Flight 3591

 - 111 Challenger & Columbia

 - 112 NASA Saturn V Documentation

 - 113 Boeing Starliner

 - 114 FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

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 - 127 Military Fratricide — Desert Storm to Afghanistan

128	Operation Eagle Claw
129	Patriot Missile / Dhahran
133	Mark 14 Torpedo Failures
134	V-22 Osprey
135	9/11 Intelligence Sharing Failures
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139	GIFT and the Adoption Gap
142	xAPI / Total Learning Architecture — Interoperability Gap
146	Northeast Blackout
151	GM Ignition Switch
152	TSB Bank IT Migration
155	Toyota Production System / Andon Cord
156	INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet
157	AI-Augmented Coding Tools
160	Davis-Besse Reactor Head Corrosion
163	Deepwater Horizon
164	Grenfell Tower
165	Camp Fire / PG&E
169	Fukushima Daiichi
174	Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed
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177	CIRAS — Confidential Incident Reporting for UK Rail
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187	COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate
190	Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment
191	Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict
192	Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model
198	Launching the BRAIN Initiative — Governance of a Grand Challenge

199	Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact
200	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver
201	Aadhaar Exclusion — Biometric Welfare Delegation and Its Judicial Reckoning in India
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205	The Discipline We Build Next
LEN 9	30 cases
2	EHR / CPOE Implementation
3	Watson for Oncology — Delegation Marketed Ahead of Capability
25	Removing the Race Coefficient from eGFR
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34	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case
35	Radiology AI Miscalibration
36	AlphaFold — Protein Structure Prediction
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback
38	iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms
46	Algorithmic Bias in Educational Predictive Analytics
47	Algorithmic Bias in Automated Exam Proctoring
67	Cognitive Tutor / Carnegie Learning
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129	Patriot Missile / Dhahran
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LEN 10

21 cases

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- 23 WHO Surgical Safety Checklist
- 39 TeamSTEPPS
- 45 Tennessee Voluntary Pre-K Study
- 53 inBloom
- 54 Summit Learning / Personalized Learning Rollout
- 60 Houston EVAAS — Value-Added Teacher Evaluation on Trial
- 133 Mark 14 Torpedo Failures
- 139 GIFT and the Adoption Gap
- 140 Navy Surface Warfare Readiness Reform
- 155 Toyota Production System / Andon Cord
- 157 AI-Augmented Coding Tools
- 164 Grenfell Tower
- 170 West Africa Ebola — The Delayed International Response
- 176 Tylenol Recall
- 179 The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed
- 205 The Discipline We Build Next

References, case by case.

Every reference cited in every case, organized by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds a **Retrieved from:** rule beneath it — a worksheet line for the reviewer to enter the canonical access path they used: a publisher URL, a DOI, an agency landing page, or a stable archival locator. The rules print blank by design, because the per-case verification is being carried out against this appendix rather than reported by it; the running status of that work is in casebook/verification-log.md. Future editions will print the locators as the review closes them out.

Style. References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

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1. Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and its recommendations, paraphrased; the pull quote is verbatim from paragraph 18 of the Final Report Summary, and differs in wording from Recommendation 155, which carries the same requirement. — Retrieved from: _____
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3. Berwick, D. (2013), A Promise to Learn — A Commitment to Act — the broader NHS-safety reform. — Retrieved from: _____
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5. NHS national clinical audit and registry documentation — extension across specialties. — Retrieved from: _____

CASE 25 Removing the Race Coefficient from eGFR

2021

1. Vyas, Eisenstein, & Jones (2020), "Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms," *NEJM* 383(9):874–882 — broader race-in-algorithms survey. — Retrieved from: _____
2. Eneanya, Yang, & Reese (2019), "Reconsidering the Consequences of Using Race to Estimate Kidney Function," *JAMA* 322(2):113–114 — the equity argument that motivated the revision. — Retrieved from: _____
3. Genzen, J. R., Souers, R. J., Pearson, L. N., et al. (2022), "Reported Awareness and Adoption of 2021 Estimated Glomerular Filtration Rate Equations Among US Clinical Laboratories, March 2022," *JAMA* 328(20):2060–2062, doi:10.1001/jama.2022.15404 — the College of American Pathologists survey: 1,124 of 3,708 responding U.S. laboratories (30.3%) had adopted the 2021 CKD-EPI creatinine equation. — Retrieved from: _____
4. Delgado et al. (2021), "A Unifying Approach for GFR Estimation: Recommendations of the NKF-ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease," *American Journal of Kidney Diseases* 79(2):268–288 (published online 2021; in print vol. 79, 2022), doi:10.1053/j.ajkd.2021.08.003. Cited by online-first year, the year of the Task Force recommendation. — Retrieved from: _____
5. Inker et al. (2021), "New Creatinine- and Cystatin C-Based Equations to Estimate GFR without Race," *New England Journal of Medicine* 385(19):1737–1749, doi:10.1056/NEJMoa2102953. — Retrieved from: _____
6. OPTN/UNOS (2022–2023), "Kidney waiting-time modifications for candidates affected by race-inclusive eGFR calculations" — the mandated backdating of Black candidates' accrued waiting time (effective January 2023). — Retrieved from: _____

CASE 26 Pulse Oximetry Across Skin Tones

1990 – 2020 – 2025

1. Jubran & Tobin (1990), "Reliability of Pulse Oximetry in Titrating Supplemental Oxygen Therapy in Ventilator-Dependent Patients," *Chest* 97(6):1420–1425 — original finding, published thirty years earlier. — Retrieved from: _____
2. Fawzy et al. (2022), "Racial and Ethnic Discrepancy in Pulse Oximetry and Delayed Identification of Treatment Eligibility Among Patients With COVID-19," *JAMA Internal Medicine* — downstream effect during the pandemic. — Retrieved from: _____
3. Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), "Racial Bias in Pulse Oximetry Measurement," *New England Journal of Medicine* 383(25):2477–2478, doi:10.1056/NEJMc2029240. — Retrieved from: _____
4. FDA (2025), "Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff,"

issued January 7 2025, Docket No. FDA–2023–N–4976; Federal Register notice 2024–31540, published January 7 2025 — the regulatory corrective-action artifact, language may evolve in final. — Retrieved from: _____

CASE 27 **Anesthesia Monitoring Standards and the APSF — The Mortality Decline** 1986 – present

1. Eichhorn, Cooper, Cullen, Maier, Philip, & Seeman (1986), “Standards for Patient Monitoring During Anesthesia at Harvard Medical School,” JAMA 256(8):1017–1020. — Retrieved from: _____
2. American Society of Anesthesiologists, “Standards for Basic Anesthetic Monitoring,” approved 21 October 1986 and amended since — the original standard and the amendment sequence that made pulse oximetry required (1989), expired-CO₂ verification of tube placement required (early 1990s), and continual capnography through a general anesthetic required (effective July 1999). — Retrieved from: _____
3. Eichhorn (1989), “Prevention of intraoperative anesthesia accidents and related severe injury through safety monitoring,” Anesthesiology 70(4):572–577. — Retrieved from: _____
4. Braz, L. G., Módolo, N. S. P., do Nascimento, P. Jr., et al. (2009), “Mortality in anesthesia: a systematic review,” Clinics (São Paulo) 64(10):999–1006 — 33 studies, 1954–2007; anaesthesia-related mortality below one death per 10,000 anaesthetics in the final two decades. — Retrieved from: _____
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6. Sjoding et al. (2020), “Racial Bias in Pulse Oximetry Measurement,” NEJM 383(25):2477–2478 — the racial-bias edge case the standard still carries (cross-reference Case 26). — Retrieved from: _____

CASE 28 **Interprofessional Education and the Evidence Gap** 2013 – 2015

1. Reeves, Perrier, Goldman, Freeth, & Zwarenstein (2013), “Interprofessional education: effects on professional practice and healthcare outcomes (update),” Cochrane Database of Systematic Reviews, doi:10.1002/14651858.CD002213.pub3. — Retrieved from: _____
2. Institute of Medicine (2015), Measuring the Impact of Interprofessional Education on Collaborative Practice and Patient Outcomes, National Academies Press, doi:10.17226/21726, NCBI Bookshelf NBK338360. — Retrieved from: _____
3. WHO (2010), Framework for Action on Interprofessional Education and Collaborative Practice — the international policy backdrop. — Retrieved from: _____
4. v2 paired cases: Team-science training (Case 40), Implementation-science training (Case 41). — Retrieved from: _____

CASE 29 **Bar-Code Medication Administration — Cue at the Point of Care** 2010

1. Poon, E. G., Keohane, C. A., Yoon, C. S., Ditmore, M., Bane, A., Levtzion-Korach, O., Moniz, T., Rothschild, J. M., Kachalia, A. B., Hayes, J., Churchill, W. W., Lipsitz, S., Whittemore, A. D., Bates, D. W., & Gandhi, T. K. (2010). Effect of bar-code technology on the safety of medication administration. New England Journal of Medicine, 362(18):1698–1707. doi:10.1056/NEJMs0907115 — the case’s primary evaluation. — Retrieved from: _____
2. Bonkowski, J., Carnes, C., Melucci, J., Mirtallo, J., Prier, B., Reichert, E., Moffatt-Bruce, S., & Weber, R. J. (2013). Effect of barcode-assisted medication administration on emergency department medication errors. Academic Emergency Medicine, 20(8):801–806 — adjacent transfer evidence. — Retrieved from: _____
3. Thompson, K. M., Swanson, K. M., Cox, D. L., Kirchner, R. B., Russell, J. J., Wermers, R. A., Storlie, C. B., Johnson, M. G., & Naessens, J. M. (2018), “Implementation of Bar-Code Medication Administration to Reduce Patient Harm,” Mayo Clinic Proceedings: Innovations, Quality & Outcomes 2(4):342–351, doi:10.1016/j.mayocpiqo.2018.09.001, PMID:30560236, PMCID:PMC6257885 — later single-site rollout reporting a 55.4% reduction in harmful medication errors (0.65 to 0.29 per 100,000 medications), by interrupted time series across fifty inpatient units. — Retrieved from: _____

4. Institute for Safe Medication Practices (2019), Guidelines for Safe Electronic Communication of Medication Information — the display and communication standards that bar-code scanning systems and eMARs are built against. — Retrieved from: _____

CASE 30 Surgical Skill Video Peer-Rating Predicts Complications

2013

1. Birkmeyer, J. D., Finks, J. F., O'Reilly, A., Oerline, M., Carlin, A. M., Nunn, A. R., Dimick, J., Banerjee, M., & Birkmeyer, N. J. O., for the Michigan Bariatric Surgery Collaborative. (2013). Surgical skill and complication rates after bariatric surgery. *New England Journal of Medicine*, 369(15):1434–1442. doi:10.1056/NEJMsa1300625 — the case's primary evaluation. — Retrieved from: _____
2. Birkmeyer, J. D., Stukel, T. A., Siewers, A. E., Goodney, P. P., Wennberg, D. E., & Lucas, F. L. (2003). Surgeon volume and operative mortality in the United States. *New England Journal of Medicine*, 349(22):2117–2127 — the volume-outcome literature the confound rests against. — Retrieved from: _____
3. Vassiliou, M. C., Feldman, L. S., Andrew, C. G., et al. (2005). A global assessment tool for evaluation of intraoperative laparoscopic skills. *American Journal of Surgery*, 190(1):107–113 — the laparoscopic skill-rating literature this video assessment sits in. — Retrieved from: _____
4. Jacobs, D. O. (2013). Cut well, sew well, do well? *New England Journal of Medicine*, 369(15):1466–1467. doi:10.1056/NEJMe1309785 — the accompanying editorial. — Retrieved from: _____

CASE 31 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

2006 – 2016

1. Vedula, Malpani, Tao, Chen, Gao, Poddar, Ahmidi, Paxton, Vidal, Khudanpur, Hager, & Chen (2016), "Analysis of the Structure of Surgical Activity for a Suturing and Knot-Tying Task," *PLOS One* 11(3):e0149174, doi:10.1371/journal.pone.0149174. — Retrieved from: _____
2. Gao, Vedula, Reiley, Ahmidi, Varadarajan, Lin, Tao, Zappella, Bejar, Yuh, Chen, Vidal, Khudanpur, & Hager (2014), "JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS): A Surgical Activity Dataset for Human Motion Modeling," *MICCAI Workshop — JIGSAWS dataset release paper*. — Retrieved from: _____
3. Reiley, Lin, Yuh, & Hager (2011), "Review of methods for objective surgical skill evaluation," *Surgical Endoscopy* 25(2):356–366 — situates the decomposition within the broader skill-assessment literature. — Retrieved from: _____
4. Birkmeyer, Finks, O'Reilly, et al. (2013), *NEJM* — the paired skill-matters study (Case 30). — Retrieved from: _____

CASE 32 Annual-Screening UI Redesign at University of Missouri Health Care

2020

1. Pierce RP, Eskridge BR, Rehard L, Ross B, Day MA, Belden JL (2020), "The Effect of Electronic Health Record Usability Redesign on Annual Screening Rates in an Ambulatory Setting," *Applied Clinical Informatics* 11(4):580–588, doi:10.1055/s-0040-1715828 — the source of record for every figure in this case. HIMSS carries a secondary write-up of the project; it describes the redesign as adding clinical decision support, which the peer-reviewed report does not, and this case follows the report. — Retrieved from: _____
2. Hussain MI, Reynolds TL, Zheng K (2019), "Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review," *JAMIA* 26(10):1141–1149, doi:10.1093/jamia/ocz095 — adjacent evidence that interaction-design changes move alert behavior. Note its outcome is prescriber acceptance of medication safety alerts, not screening completion; it is background, and no figure in this case rests on it. — Retrieved from: _____
3. Ratwani RM, Reider J, Singh H (2019), "A decade of health information technology usability challenges and the path forward," *JAMA* 321(8):743–744, doi:10.1001/jama.2019.0161 — the call for lifecycle evaluation "from initial product design to local implementation followed by periodic reassessment," and the framing for the site-configuration gap in federal certification requirements. — Retrieved from: _____

4. Middleton et al. (2013), "Enhancing patient safety and quality of care by improving the usability of electronic health record systems," JAMIA 20(e1):e2–e8 — the framing peer-reviewed paper on EHR-usability-as-safety. — Retrieved from: _____

CASE 33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal 2019 – 2024

1. Ancker et al. (2017), "Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system," BMC Medical Informatics and Decision Making 17:36 — adjacent measurement evidence. — Retrieved from: _____

2. Hussain, M. I., Reynolds, T. L., & Zheng, K. (2019), "Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review," JAMIA 26(10):1141–1149, doi:10.1093/jamia/ocz095 — thirty-nine studies; interruptive modals accepted least often; role tailoring the only alternative that appeared to increase prescriber acceptance. — Retrieved from: _____

3. Fallon A, Haralambides K, Mazzillo J, Gleber C (2024), "Addressing Alert Fatigue by Replacing a Burdensome Interruptive Alert with Passive Clinical Decision Support," Applied Clinical Informatics 15(1):101–110, doi:10.1055/a-2226-8144 — 80% cut in alert firings; COVID precautions ordered rose from 23% to 61%. — Retrieved from: _____

4. Van Dort, B. A., Zheng, W. Y., Sundar, V., & Baysari, M. T. (2021), "Optimizing clinical decision support alerts in electronic medical records: a systematic review of reported strategies adopted by hospitals," JAMIA 28(1):177–183, doi:10.1093/jamia/ocaa279 — eight published optimization programs; six evaluated their changes and five of those reported a reduction in alert rate. — Retrieved from: _____

5. Office of the National Coordinator for Health IT, SAFER Guides on CDS — practitioner-tier guidance the redesigns instantiate. — Retrieved from: _____

6. Patterson E (2024), "Navigating Alert Fatigue: A Case Study in Electronic Health Record Alert Design Optimization," Studies in Health Technology and Informatics 315:447–451, doi:10.3233/SHTI240188 — nursing-workflow QI redesign of four high-firing alerts; all four removed, 877 unactionable disruptive nursing hours released. — Retrieved from: _____

CASE 34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case 2022 – 2024

1. Boussina, A., Shashikumar, S. P., Malhotra, A., Owens, R. L., El-Kareh, R., Longhurst, C. A., Quintero, K., et al. (2024), "Impact of a deep learning sepsis prediction model on quality of care and survival," NPJ Digital Medicine 7:14, doi:10.1038/s41746-023-00986-6. — Retrieved from: _____

2. Shashikumar, S. P., Wardi, G., Malhotra, A., & Nemati, S. (2021), "Artificial intelligence sepsis prediction algorithm learns to say 'I don't know,'" NPJ Digital Medicine 4:134 — the methodological-foundation paper for the conformal-prediction abstention structure. — Retrieved from: _____

3. Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070 — the external-validation paper on Epic Sepsis (Case 5). — Retrieved from: _____

4. Adams, R., Henry, K. E., Sridharan, A., Soleimani, H., Zhan, A., Rawat, N., Johnson, L., et al. (2022), "Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis," Nature Medicine 28:1455–1460 — the prospective-positive TREWS deployment paper (Case 20). — Retrieved from: _____

CASE 35 Radiology AI Miscalibration 2018 – present

1. Wu, E., Wu, K., Daneshjou, R., Ouyang, D., Ho, D. E., & Zou, J. (2021), "How medical AI devices are evaluated: limitations and recommendations from an analysis of FDA approvals," Nature Medicine 27(4):582–584, doi:10.1038/s41591-021-01312-x — the 130-device audit (126 retrospective-only, none of the 54 high-risk devices prospective, 93 without publicly reported multi-site assessment, 17 reporting demographic subgroup performance) and the cross-site pneumothorax experiment behind this case's diagram. — Retrieved from: _____

2. Seyyed-Kalantari, L., Zhang, H., McDermott, M. B. A., Chen, I. Y., & Ghassemi, M. (2021), "Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations," *Nature Medicine* 27:2176–2182 — disparities across Black, Hispanic, female, and lower-socioeconomic subgroups; underdiagnosis highest for intersectional subpopulations such as Hispanic female patients. — Retrieved from: _____
3. Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," *Science* 366(6464):447–453 — proxy-label bias in a care-management algorithm; 200 million Americans/year are subject to tools of this class. — Retrieved from: _____
4. Daneshjou, R., Vodrahalli, K., Novoa, R. A., et al. (2022), "Disparities in dermatology AI performance on a diverse, curated clinical image set," *Science Advances* 8(32):eabq6147, doi:10.1126/sciadv.abq6147 — the Diverse Dermatology Images set; models and labeling dermatologists both degrade on dark skin tones. — Retrieved from: _____
5. FDA, "Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device" (2019); FDA's public "Artificial Intelligence-Enabled Medical Devices" authorization list; final Predetermined Change Control Plan guidance (December 2024) and the draft "Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations" (January 2025), with Good Machine Learning Practice principles — the regulatory trajectory; absence of mandatory demographic stratification at clearance and post-market monitoring of in-use performance. — Retrieved from: _____
6. Larrazabal, A. J., Nieto, N., Peterson, V., Milone, D. H., & Ferrante, E. (2020), "Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis," *PNAS* 117(23):12592–12594 — AUC drops for under-represented genders on NIH ChestX-ray14 and CheXpert, across three network architectures. — Retrieved from: _____
7. Bernhardt, M., Jones, C., & Glocker, B. (2022), "Potential sources of dataset bias complicate investigation of under-diagnosis by machine learning algorithms," *Nature Medicine* 28(6):1157–1158 — one of three Matters Arising contesting Seyyed-Kalantari et al.; the authors' reply is at 28(6):1161–1162, doi:10.1038/s41591-022-01854-8. Cited here so the contested status travels with the finding. — Retrieved from: _____
8. Wachter, R. M. & Brynjolfsson, E. (2024), "Will Generative Artificial Intelligence Deliver on Its Promise in Health Care?" *JAMA* 331(1):65–69, doi:10.1001/jama.2023.25054 — the counterweight argument that health care is readier for generative AI than for its predecessors and the lag to benefit may be shorter. It is not a bias or evaluation paper; do not cite it for the proxy-and-population problem. — Retrieved from: _____

CASE 36 AlphaFold — Protein Structure Prediction

2020 – present

1. Jumper, J., Evans, R., Pritzel, A., et al. (2021), "Highly accurate protein structure prediction with AlphaFold," *Nature* 596(7873):583–589, doi:10.1038/s41586-021-03819-2 — the method, and the CASP14 validation. — Retrieved from: _____
2. Varadi, M., Bertoni, D., Magana, P., et al. (2024), "AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences," *Nucleic Acids Research* 52(D1):D368–D375, doi:10.1093/nar/gkad1011 — the database at 214M, up from 300k in 2021, and its integration into PDB, UniProt, Ensembl and InterPro. — Retrieved from: _____
3. CASP14 press release, Protein Structure Prediction Center, 30 November 2020 — GDT above 90 on roughly two-thirds of targets. Note the Moulton quotation this case carries is **not** in the press release: it is from Google DeepMind's post of the same date, "AlphaFold: a solution to a 50-year-old grand challenge in biology," which also reports the 92.4 median GDT. — Retrieved from: _____
4. Tunyasuvunakool, K., Adler, J., Wu, Z., et al. (2021), "Highly accurate protein structure prediction for the human proteome," *Nature* 596(7873):590–596, doi:10.1038/s41586-021-03828-1 — 98.5% of human proteins covered; 58% of residues confidently predicted, 36% of all residues at very high confidence, against 17% of residues covered by experimental structure beforehand. — Retrieved from: _____
5. AlphaFold 3 access chronology: publication with neither code nor weights (May 2024), the open letter signed by more than a thousand researchers, the *Nature* editorial "AlphaFold3 — why did *Nature* publish it without its code?" (2024, doi:10.1038/d41586-024-01463-0), and the 11 November 2024 GitHub release of the code under a non-commercial license with weights by application — the open-release governance decision as actually written. — Retrieved from: _____

6. Abramson, J., Adler, J., Dunger, J., et al. (2024), "Accurate structure prediction of biomolecular interactions with AlphaFold 3," Nature 630(8016):493–500, doi:10.1038/s41586-024-07487-w; and the 2024 Nobel Prize in Chemistry, one half to David Baker for computational protein design and one half jointly to Demis Hassabis and John M. Jumper for protein structure prediction. — Retrieved from: _____

CASE 37 DeepMind Mammography — High-Profile Nature Paper, Replicability Push-back 2020

1. McKinney, S. M., Sieniek, M., Godbole, V., Godwin, J., Antropova, N., Ashrafi, H., Back, T., et al. (2020), "International evaluation of an AI system for breast cancer screening," Nature 577:89–94, doi:10.1038/s41586-019-1799-6. — Retrieved from: _____

2. Haibe-Kains, B., Adam, G. A., Hosny, A., Khodakarami, F., MAQC Society Board of Directors, Waldron, L., Wang, B., et al. (2020), "Transparency and reproducibility in artificial intelligence," Nature 586:E14–E16, doi:10.1038/s41586-020-2766-y. — Retrieved from: _____

3. McKinney, S. M., Karthikesalingam, A., Tse, D., Kelly, C. J., Liu, Y., Corrado, G. S., & Shetty, S. (2020), "Reply to: Transparency and reproducibility in artificial intelligence," Nature 586:E17–E18, doi:10.1038/s41586-020-2767-x — the developers' response, published alongside the addendum below. — Retrieved from: _____

4. McKinney et al. (2020), "Addendum: International evaluation of an AI system for breast cancer screening," Nature 586:E19, doi:10.1038/s41586-020-2679-9 — published in the same issue as the Haibe-Kains comment; expands the supplementary methods with optimization hyperparameters and the augmentation strategy to assist replication. Code and trained models were not released. — Retrieved from: _____

5. Freeman, K., Geppert, J., Stinton, C., Todkill, D., Johnson, S., Clarke, A., & Taylor-Phillips, S. (2021), "Use of artificial intelligence for image analysis in breast cancer screening programmes: systematic review of test accuracy," BMJ 374:n1872, doi:10.1136/bmj.n1872 — twelve studies, 131,822 women, McKinney included; no prospective studies found, studies of poor methodological quality, 34 of 36 AI systems less accurate than a single radiologist and all less accurate than two in consensus. — Retrieved from: _____

6. MASAI trial (Lång et al.): clinical safety analysis, Lancet Oncology 2023;24(8):936–944, doi:10.1016/S1470-2045(23)00298-X; Hernström et al., screening performance, Lancet Digital Health 2025;7(3):e175–e183, doi:10.1016/S2589-7500(24)00267-X; Gommers et al., interval cancer, sensitivity and specificity, The Lancet 2026;407(10527):505–514, doi:10.1016/S0140-6736(25)02464-X. 105,934 women randomized; the AI system is Transpara v1.7.0 (ScreenPoint Medical), not the system evaluated by McKinney et al. — Retrieved from: _____

CASE 38 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011

1. FDA, iPLEDGE program documentation (2006 – present) — REMS architecture and enrollment requirements. — Retrieved from: _____

2. Sullivan et al. (2003), House Science Committee statement on SUBSAFE — the structural counterpoint (Case 173). — Retrieved from: _____

3. Tkachenko, E., Singer, S., Sharma, P., Barbieri, J., & Mostaghimi, A. (2019), "US Food and Drug Administration reports of pregnancy and pregnancy-related adverse events associated with isotretinoin," JAMA Dermatology, 155(10):1175–1179 — source of the 218 to 310 annual reports after 2011. — Retrieved from: _____

4. Shin, J., Cheetham, T. C., Wong, L., Niu, F., Kass, E., Yoshinaga, M. A., Sorel, M., McCombs, J. S., & Sidney, S. (2011), "The impact of the iPLEDGE program on isotretinoin fetal exposure in an integrated health care system," Journal of the American Academy of Dermatology, PMID:21565419. — Retrieved from: _____

5. Collins, M. K., Moreau, J. F., Opel, D., Swan, J., Prevost, N., Hastings, M., Schwarz, E. B., & Ferris, L. K. (2014), "Compliance with pregnancy prevention measures during isotretinoin therapy," Journal of the American Academy of Dermatology, 70(1):55–59, PMID:24157382 — source of the 150 annual exposures figure. — Retrieved from: _____

CASE 39 TeamSTEPPS

2006 – present

1. Salas, E., DiazGranados, D., Klein, C., Burke, C. S., Stagl, K. C., Goodwin, G. F. & Halpin, S. M. (2008), "Does Team Training Improve Team Performance? A Meta-Analysis," *Human Factors* 50(6):903–933 — the cross-domain team-training evidence base. — Retrieved from: _____
2. Umscheid, C. & Haugstetter, M. (2023), "With TeamSTEPPS 3.0, AHRQ Refreshes a Landmark Patient Safety Training Curriculum," *AHRQ Views* (12 September) — the DoD partnership and the first-eight-years adoption figures (1,500 hospitals, 5,000 master trainers, 300,000 professionals). — Retrieved from: _____
3. Weaver, S. J., Dy, S. M. & Rosen, M. A. (2014), "Team-training in healthcare: a narrative synthesis of the literature," *BMJ Quality & Safety* 23(5):359–372 — bundled team-training interventions, with tools and organisational change to support transfer, show the largest effects. — Retrieved from: _____
4. Weld, L. R. et al. (2016), "TeamSTEPPS Improves Operating Room Efficiency and Patient Safety," *American Journal of Medical Quality* 31(5):408–414 — the on-time first-start figure. — Retrieved from: _____
5. AHRQ, TeamSTEPPS 3.0 Curriculum (2023) — the framework and four competencies. — Retrieved from: _____

CASE 40 Team Science Training for Clinical and Translational Scientists

2020 – 2022

1. Mook, A., Knerich, V., Komaie, G., Cicutto, L., & Cross, J. (2025), "Team science training for clinical and translational scientists: An assessment of effectiveness," *Journal of Clinical and Translational Science* 9(1):e158, doi:10.1017/cts.2025.10088. — Retrieved from: _____
2. Falk-Krzesinski et al. (2011), "Mapping a research agenda for the science of team science," *Research Evaluation* 20(2):143–156 — broader team-science literature backdrop. — Retrieved from: _____
3. National Research Council (2015), *Enhancing the Effectiveness of Team Science* (National Academies Press) — the National Academies team-science synthesis. — Retrieved from: _____
4. Paired cases in this volume: IPE evidence gap (Case 28), Implementation Science Training (Case 41) — the frontier/ measurement companions. — Retrieved from: _____

CASE 41 Translational-Workforce Training — Stated Goals Outrunning Named Practice

2020s

1. Morris, Wooding, & Grant (2011), "The answer is 17 years, what is the question: understanding time lags in translational research," *Journal of the Royal Society of Medicine* — the original 17-year-gap source for Case 13. — Retrieved from: _____
2. Sancheznieto, F., Sorkness, C. A., Attia, J., et al. (2021), "Clinical and translational science award T32/TL1 training programs: program goals and mentorship practices," *Journal of Clinical and Translational Science* 6(1):e13, doi:10.1017/cts.2021.884. — Retrieved from: _____
3. Brownson, Colditz, & Proctor (2018), *Dissemination and Implementation Research in Health* (2nd ed.) — the broader implementation-science synthesis. — Retrieved from: _____
4. Paired cases in this volume: Team-science training (Case 40), IPE evidence gap (Case 28). — Retrieved from: _____

CASE 42 Australian Hospital-Pharmacy Technician Role Redesign

2016

1. Pharmacy Board of Australia and SHPA practice statements limiting technician work to activities not requiring professional judgement — the regulatory framing the project sits inside. — Retrieved from: _____
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2014

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Across U.S. Higher Education

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USC × 2U Online MSW — When the Delegation Becomes the Product
(Luna v. USC)

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2010

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CASE 67 Cognitive Tutor / Carnegie Learning

1990s – present

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2017 – 2023 (six cycles)

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